

Statistical Modeling

DATA.STAT.740

Henri Pesonen 2025-2026

Contact

Course is in Moodle - please use **General discussion**

Course updates are posted to **News**

You can contact me via henri.pesonen@tuni.fi

If you remember, start the message with “Statistical Modeling” - I’ll reserve time for going through the course correspondence

Office TC336 - stop by or send an email before to make sure I’m there

Contact for Master’s theses topics

Pass the final exam - pass the course

Course consists of

- Lectures

- Course notes/lecture slides

- Weekly exercises

- Supplementary reading

Lectures

Not mandatory but recommended

Lecture material of the week will be online on Monday every week

Go over the material **before** the lectures (this is the way)

We will focus on practical material, examples and discussion on the lectures

There will be a recording available of the lectures

Same material - content may vary

Exercises

Not mandatory but recommended

Solutions can be returned via Moodle

Earn up to 6 points for final exam - max 1 point from each exercise

Example

A student earns 4 exercise points from the first exercise out of 6 exercise points available in the exercise session

For the exam they'll receive 4/6 points from that exercise session.

Maximum exercise points from weekly exercises may vary.

Grading

Accumulated points	Grade
[0, 7) (just the exam points)	Failed
[7, 13)	1
[13, 19)	2
[19, 25)	3
[25, 31)	4
[31, 42]	5

Exercises

You can come to the exercise sessions on Fridays 14-16 (K1241)

The first session is 9.1.

If you want to return exercises via Moodle, the deadline is on Fridays at 14:00

The deadline for the first exercise is 16.1. at 14:00

No exercise session 16.1. - the 2nd exercise session is 23.1. and the Moodle DL is also 23.1.

You can work alone or together with other students on the problems and I will be on the location to give feedback, hints and solutions.

Exercises

First two questions are same every week

Problem 1 (0.5 points)

If you have made the problems with your co-students, please list their names as the solution to this problem or “Alone” - if you're at the exercise session, this is not necessary. There is no penalty for working together, and is even encouraged, but I'd like to know who worked together, so I don't need review each solution separately. You will receive this point if you have attempted to answer any problems beyond the first two problems. **Communication with co-students is not possible in the exam.**

Exercises

First two questions are same every week

Problem 2 (0.5 points)

Did you use a large language model (LLM) such as ChatGPT to help with these assignments? Please list which model(s) you used and how you used them. If you did not use any LLM, please state that by writing “No LLMs”. There are no penalties for using LLMs, but I'd like to know who used them and how. Your feedback will be dependent on this information. **LLMs are unavailable in the exam.**

LLMs - The less you know, the more you think you know

Please use responsibly

Be aware that credible looking solutions may not be correct

Without some knowledge, you don't know what you don't know
Don't become a Dunning-Kruger-machine

You'll be able to pass the course without faking the knowledge
Don't come to the exam without studying yourself

Exam

TUNI EXAM

A mixture of statistical programming and theoretical questions

More info in a few weeks

Grading

Accumulated points	Grade
[0, 7) (just the exam points)	Failed
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Statistical software



Not a great language for general purpose programming

Is a great language for statistical modeling and data science as it has been **developed for statistical analysis**

A large number of helpful libraries accessible from CRAN

Prerequisites

Basic matrix calculus

Continuous and discrete random variables and probability distributions

The mean and the variance of a random variable

Hypothesis testing and confidence intervals

Main topics

Statistical modeling loop

Generalized linear models

Maximum likelihood theory

Regularized linear regression

Survival analysis

Model fit diagnostics

Generalized linear models

The basic theory of generalized linear models

The most common generalized linear models and when to use and when to not use them

Which predictors to include into the model

Data transformations

Maximum likelihood theory

Exponential family of distributions

How to fit different models using maximum likelihood theory

How calculate and interpret parameter and confidence interval estimates

Regularized linear regression

What is it and when to use it?

Different regularization methods

Automatic predictor selection

Survival analysis

Time-to-event-data

Kaplan-Meier plots

Cox regression

Model fit diagnostics

Parameter testing

Model comparison with likelihood-ratio test

Goodness-of-fit numbers

What should you know how to do after taking this course?

Understand the statistical question

Explore the data with visuals and summaries and identify initial patterns before modeling

Select and construct a statistical model with justification

Explain what do the results mean in the context of the original problem, and what are the limitations of the model?

Justify whether the model fits the data or should we revise it?

Statistical Modeling

DATA.STAT.740

Lecture 1

Henri Pesonen 2025-2026

Intro

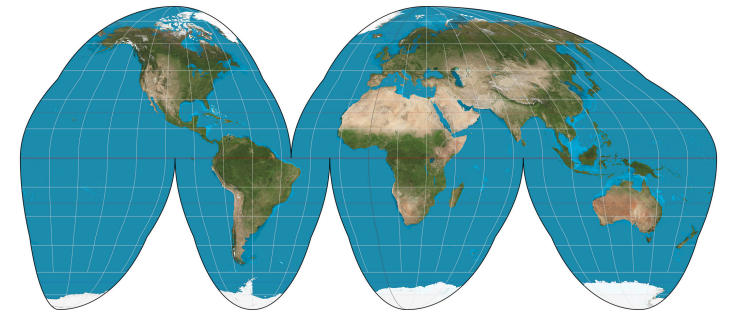
Models are important for learning how the world works



Prototype inventions



Architectural small-scale
mock-ups



Geographical maps

Models are used to

Find explanations

Predict outcomes

Test assumptions

Make comparisons between the effects of different modeling assumptions

In the world of empirical data, we use **statistical modeling**

A mathematical framework for inferring knowledge about the world using observations

Essential piece of modern **data analysis**

Observations can be **empirically collected** or **simulated**

At the heart of statistical modeling are **statistical models**

Statistical modeling is used to

Understand trends and patterns

Identify which variables are related to a response variable and how

Make predictions about future observations

Separate signal from noise

Understand underlying, unexplained variability in the patterns

Compare different models to each other.

Statistical models

Describe mechanisms and the uncertainty how data are related to each other

Models describe a **data generating process**

The outcome of a model relies on probability distributions

Probability theory is the way to describing the uncertainty of the outcome of the data generating process

Supervised and unsupervised learning

Statistical modeling can be roughly divided into supervised and unsupervised learning (also semi-supervised learning etc.)

Supervised learning

Uses training data with correct labels

Unsupervised learning

Learn patterns from unlabeled data.

Statistical modeling loop

Statistical modeling is **not** really about finding a one model - **the truth**

It's a creative process of constructing and fitting (multiple) models

“All models are wrong, but some are useful”
- George Box

Statistical modeling loop

1. Build statistical models based on the information and assumptions about the system
2. Fit the statistical models using empirical data. Usually this means estimating **parameters** of the models that control the probabilistic outputs of the models
3. Diagnose and critique the models. Evaluate how well the model fits to the data. Can the models be used to predict future outcomes? Identify models' strengths and weaknesses. If you have competing models which is the best?
4. Loop back to the building stage, if the models are unsatisfactory. In some cases, determine that you need more and/or better empirical data!

Different types of observations

The starting point of building a statistical model is to figure out what type of data you have

Data are measured in different types of **scales**

Scales determines what models are reasonable

From the scales it's important to determine

What are possible values of an observation

Is the scale **nominal, ordinal, interval or ratio?**

Nominal scale

Qualitative measurements, e.g.

hospital from where the datapoint is recorded

the birth country of a patient

We can compare the equality of two datapoints, but can't order them naturally

if $y_1 = \text{TAYS}$
 $y_2 = \text{TYKS}$ then $y_1 \neq y_2$ is true but we can't say $y_1 < y_2$ or $y_1 > y_2$

Ordinal scale

Qualitative measurements

Difference to the nominal scale is that there is a natural ordering of possible values

Likert-scale example: y_1 = strongly agree
 y_2 = agree
 y_3 = neutral
 y_4 = disagree
 y_5 = strongly disagree

We can say $y_1 < y_2 < y_3 < y_4 < y_5$ but the difference between y_1 and y_2 is not the same as between y_2 and y_3

Interval scale

Quantitative measurements

We can order the measurements, add and subtract measurement and compare lengths of intervals

Interval scales contain an arbitrary zero point

Measurement 0 is not the complete absence of something

Example

The temperature measured in Celsius degrees 10°C is **not** twice 5°C and 0°C is not **the absence of temperature**

Ratio scale

Quantitative measurements

Similar to the interval scale

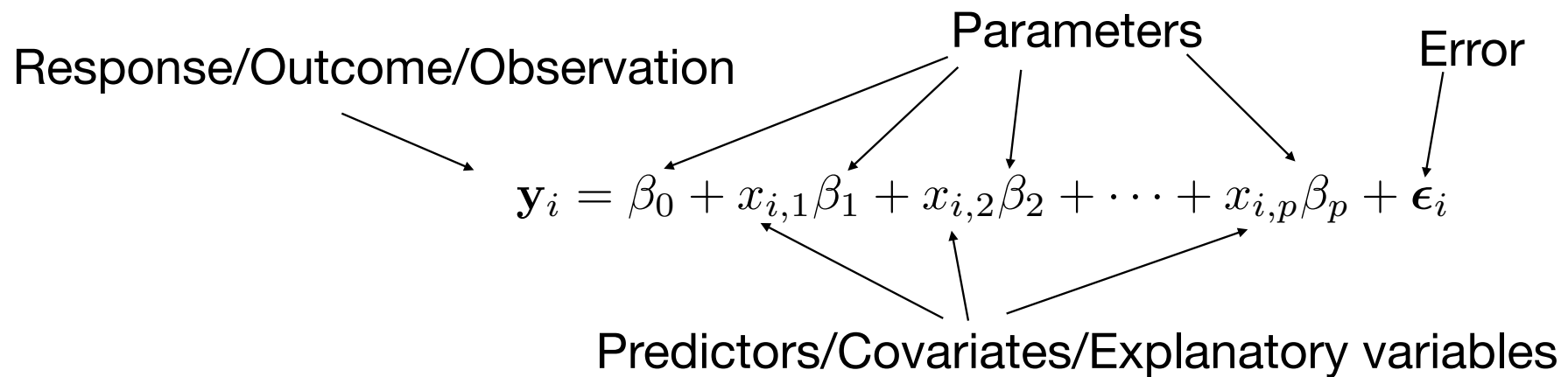
The main difference is that in the ratio scale, there is a true zero point, and negative numbers are impossible.

The weight and height are examples of variables measured in the ratio scale, that is 2kg is twice the 1kg mass and 0kg is the absence of mass

Generative linear models

Normal linear model

A simple generalized linear model that will be our starting point



Normal linear model

Usually the model is written using linear algebra

$$\begin{aligned} \mathbf{y}_i &= \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \cdots + x_{i,p}\beta_p + \boldsymbol{\epsilon}_i \\ &= \mathbf{x}_i^T \boldsymbol{\beta} + \boldsymbol{\epsilon}_i \end{aligned}$$

where $\mathbf{x}_i^T = [1 \quad x_{i,1} \quad \cdots \quad x_{i,p}]$

$$\boldsymbol{\beta}^T = [\beta_0 \quad \beta_1 \quad \cdots \quad \beta_p]$$

Normal linear model

When there are N observations from the model $\mathbf{y}_i = x_i^T \beta + \epsilon_i$:

$$\mathbf{y} = X\beta + \epsilon$$

where

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_N \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ 1 & x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & x_{N,1} & x_{N,2} & \cdots & x_{N,p} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_N \end{bmatrix}$$

Design matrix/Independent variables matrix/Feature matrix

Random outcome

Notice the bold font letters in $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$

We write random variables in bold font

Outcome $\mathbf{y}$ is random because the error $\boldsymbol{\epsilon}$ is random

Normal linear model $\epsilon_i \stackrel{\text{i.i.d}}{\sim} \text{Normal}(0, \sigma^2)$

normal distribution with mean 0 and variance σ^2

independent and identically distributed

this means also that the errors
are *homoscedastic* as opposed
to *heteroscedastic*

Normal linear model

Because the error components are i.i.d: $\epsilon \sim \text{Normal}(0, \sigma^2 I_N)$

identity matrix $I_N \in \mathbb{R}^{N \times N}$



The probability distribution of the observation is

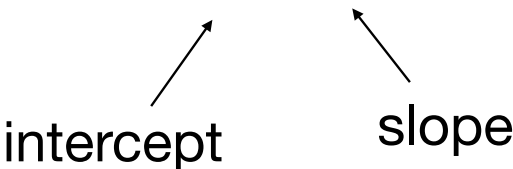
$$p(y \mid X, \beta, \sigma) = \text{Normal}(X\beta, \sigma^2 I).$$

Remember that for normal distributed multivariate random variable

$\mathbf{x} \sim \text{Normal}(\mu, \Sigma)$ and for any non-random matrix A and vector c ,
its affine transformation is $A\mathbf{x} + c \sim \text{Normal}(A\mu + c, A\Sigma A^T)$

Linear regression with a single continuous predictor

Normal linear model is a straight line in a two-dimensional space

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$


intercept slope

Intercept : What is the expected observation when the predictor is 0

Slope : How much does the observation change when the predictor changes by one measurement unit?

At each predictor value the observation is distributed as $\text{Normal}(\beta_0 + \beta_1 x_i, \sigma^2 I)$

Example: Volume of black cherry tree timber

This dataset (blackcherry.txt) includes the measurements of the girth, height and volume of timber in felled black cherry trees

Girth is the diameter of the tree (in inches) measured at 4ft6in above the ground, height is measured in ft and volume of timber is measured in cubic ft

For now, we'll use only girth as the predictor of the volume of timber



Linear regression with a single categorical predictor

Let the predictor have two possible values $x_i \in \{\text{Group1}, \text{Group2}\}$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Normal linear model is now a set of normal distributions

Let's use indicator or dummy coded variables by selecting $x_i = \text{Group1}$ as the reference and introducing

$$x_{i,2} = \begin{cases} 0, & \text{if } x_i \in \text{Group1} \\ 1, & \text{if } x_i \in \text{Group2} \end{cases}$$

Linear regression with a single categorical predictor

Using the dummy variable, the model becomes

$$y_i = \beta_0 + \beta_1 x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

or

$$y_i = \begin{cases} \beta_0, & \text{if } x_i \in \text{Group1} \\ \beta_0 + \beta_1, & \text{if } x_i \in \text{Group2} \end{cases}, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

where β_0 the common intercept for both groups and β_1 is the difference between the group means!

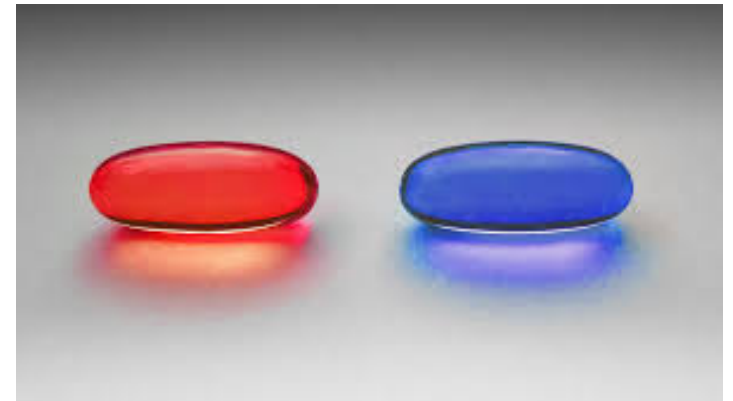
Difference between groups is often interesting quantity

Example: Placebo vs Treatment

In clinical trials a basic setup is that a new drug has been developed and we wish to test its effectiveness in treating disease Z by eliminating the effect of chance in the treatment outcomes. Two groups consisting of patients with disease Z that are statistically indistinguishable from one another have been randomly allocated one of two groups.

Patients in group A will receive a sugar pill

Patients in group B will receive a dose of new drug



Linear regression with a single categorical predictor

Let the predictor have K possible values $x_i \in \{\text{Group1}, \dots, \text{GroupK}\}$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Dummy coded variables

	x_i^1	x_i^2	$\dots$	x_i^K
Group1	1	0	$\dots$	0
Group2	0	1	$\dots$	0
$\vdots$				
GroupK	0	0	$\dots$	1

Linear regression with a single categorical predictor

Dummy coded variables

	x_i^1	x_i^2	$\dots$	x_i^K
Group1	1	0	$\dots$	0
Group2	0	1	$\dots$	0
$\vdots$				
GroupK	0	0	$\dots$	1

Because exactly one of the dummy variables is always 1 the rest is 0, we select one of the groups as the reference Group

Linear regression with a single categorical predictor

The models for each group becomes

$$\begin{aligned}x_i = \text{Group1} : \quad y_i &= \beta_0 + \epsilon_i, \\x_i = \text{Group2} : \quad y_i &= \beta_0 + \beta_1^2 + \epsilon_i \\&\vdots \\x_i = \text{GroupK} : \quad y_i &= \beta_0 + \beta_1^K + \epsilon_i,\end{aligned}$$

Each group models have a common intercept and a coefficient indicating the difference of the average response with respect to the reference group average response

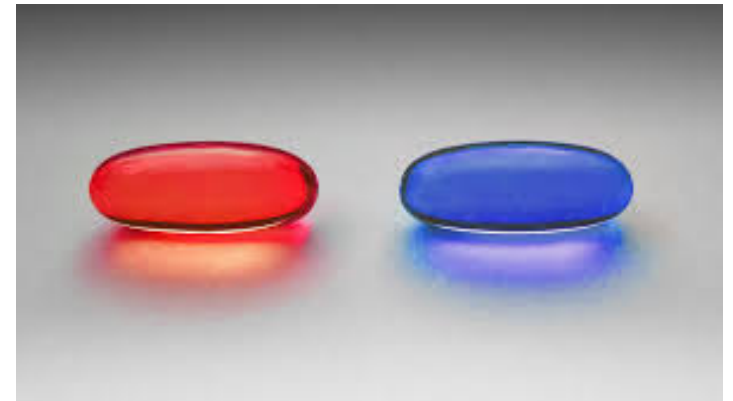
Linear regression with a single categorical predictor

The recoded linear normal model is $y = X\beta + \epsilon$ where

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{matrix} \text{Group1} \\ \\ \text{Group2} \\ \\ \text{Group3} \\ \\ \\ \text{GroupK} \end{matrix}$$

Example: multi group comparison

The comparison of outcomes from multiple groups is a standard statistical modeling problem. Let there be five groups of 10 students each taking a standardized test. Let's investigate how the data looks like when interpreted in different ways.



End of lecture 1